THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is NOT a heating-effect appliance? 1

(a) toaster **(b)** geyser **(c)** iron **(d)** electric bell

2. A circuit breaker is better than a fuse because it can be: 1

(a) melted **(b)** reset after tripping

(c) thrown away **(d)** made of tungsten

3. If the live wire touches the metal body of an appliance and a person touches it, he may get: 1

(a) a fever **(b)** cured **(c)** a cold **(d)** a shock

4. When current flows in the coil on an iron core, the iron: 1

(a) melts **(b)** becomes an insulator

(c) gets magnetised **(d)** becomes a conductor

5. Electric current is flowing in a straight wire from south to north. The compass needle placed above it is: 1

(a) demagnetised **(b)** undisturbed **(c)** melted **(d)** deflected

6. What would you choose as the best material for a fuse wire? 1

(a) tin-lead alloy **(b)** iron **(c)** nichrome **(d)** tungsten

7. A student reverses the cell connections in an electromagnet. The poles of the electromagnet: 1

(a) disappear **(b)** stay the same

(c) get interchanged **(d)** become weaker permanently

8. A student winds the wire on an iron nail without insulation (bare wire). The electromagnet will: 1

(a) work better

(b) melt the nail

(c) not work well because the current short-circuits

(d) reverse

9. A student's circuit has two bulbs in series. If one bulb fuses, the other will: 1

(a) glow brighter **(b)** be unaffected **(c)** blow too **(d)** also go off

10. Soft iron is preferred for the core because it: 1

(a) is heavy

(b) conducts heat

(c) is cheap only

(d) loses magnetism quickly when current stops

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A battery of three cells is made by connecting them in series. Why is the positive terminal of one joined to the negative of the next? 2

12. How is the direction of deflection of the compass needle related to the direction of current? 2

13. Why is it dangerous to use a fuse wire of higher rating than needed? 2

14. Why are scrap yards using electromagnets and not permanent magnets? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the role of the iron core inside an electromagnet. What would happen without it? 3

16. State two precautions to be observed while using a compass in Oersted's experiment. 3

17. How does the thickness of the wire affect the strength of an electromagnet? Give one reason. 3

OR

Compare the heating of a fuse wire with that of a copper wire carrying the same current. 3

18. A student placed a nail inside a coil; the nail stuck to pins when the current flowed and dropped them when it was off. Name the device and explain. 3

OR

Why do electricians wear rubber gloves and use tools with insulating handles? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain why nichrome is used in heater coils, tungsten in bulb filaments and a tin-lead alloy in fuse wire, referring to properties. 5

OR

A power strip has 5 sockets. Why is it dangerous to run an iron, a geyser, a heater, a TV and a mixer from one strip? Suggest a solution. 5

20. Describe the process to make a stronger electromagnet and explain each factor: core, turns, current. 5

OR

Explain the working of an electric iron. Name the parts and the effect of current used. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND